

Jacob Jerry Arackal

AI Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Dev.to](#)

Summary

AI Engineer focused on building production-ready GenAI systems, RAG pipelines, AI agents, and MLOps workflows. Specialized in LLMOps, Agentic AI, Machine Learning, and Full-Stack Development. Passionate about shipping reliable AI systems and sharing technical learnings through engineering content.

Professional Experience

Techfriar Technologies

AI-Native Full-Stack Engineer | May 2024 – Mar 2025

- Architected and deployed RAG pipelines and LLM agents using Gemini, OpenAI, and Llama models.
- Built AI workflow automation systems using LangChain and Agentic AI patterns.
- Developed scalable full-stack applications using React, Next.js, GraphQL, PostgreSQL, and Python.
- Containerized applications using Docker, ensuring consistent development and production environments.
- Optimized backend systems and database performance for enterprise applications.

Microsoft

AI/ML Engineer Intern | Jan 2022 – May 2022

- Built an automobile price prediction model using Python and machine learning techniques.
- Leveraged Azure ML services for experimentation and deployment.
- Performed feature engineering, model evaluation, and hyperparameter tuning.

Key Projects

Harness Engineering — Production-Grade AI Agent Harness

- Built a production-oriented **agent harness** for controlled execution, tool access, context management, and validation of AI agents.
- Designed execution and safety mechanisms separating **agent reasoning from infrastructure, tools, and environment control**.
- Focused on reliability, reproducibility, failure handling, and observability for production agent workflows.

MemoryOps AI — Context & Memory Infrastructure

- Built a context and memory infrastructure for maintaining **session, workspace, and persistent memory** across agent executions.
- Designed context retrieval and state hydration mechanisms to provide agents with relevant information across multi-step workflows.
- Explored **context engineering as an infrastructure layer** for reliable long-running AI agents.

EvalForge — AI Evaluation Framework

- Built an evaluation framework for systematically testing **LLM and agent behavior** using datasets, test cases, experiments, and metrics.
- Designed repeatable evaluation workflows to compare agent performance across experiments instead of relying solely on qualitative inspection.
- Focused on making AI evaluation **measurable, reproducible, and integrated into the engineering lifecycle**.

Multi-Agent Workflow Orchestrator

- Built a multi-agent orchestration platform for coordinating **specialized agents across complex workflows**.
- Implemented workflow execution, agent coordination, context propagation, state management, and execution lifecycle handling.
- Designed the system around production concerns including **fault isolation, reliability,**

Technical Skills

- GenAI & Agentic AI: LangChain, LangGraph, RAG, AI Agents, Multi-Agent Systems, MCP, LLMops, Prompt Engineering
- Machine Learning: Scikit-Learn, XGBoost, Random Forest, Logistic Regression, Feature Engineering, Model Evaluation
- MLOps & Cloud: MLflow, Docker, GitHub Actions, CI/CD, FastAPI, Model Serving, Experiment Tracking
- Programming: Python, JavaScript, TypeScript, SQL, Node.js
- Databases: PostgreSQL, MongoDB, Pinecone, ChromaDB, SQLite, FAISS
- Frontend: React.js, Next.js, HTML, CSS

Education

Bachelor of Technology (Information Technology) | Amal Jyothi College of Engineering |
2019 – 2023

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional